

PROJECT DETAILS

Level 3 can be projected as : **Communication chip**

Level 4 can be projected as : **Navigation chip**

using sanchain report + scan cell report info.

CORE A:

no. of int mode chains = 15 (core logic ffs of core A + wrapper ffs of core A)
no. of ffs in each chain = 45

$$\text{total flops in core A} = \frac{45}{\times 15} \\ \underline{\hspace{1cm}} 675 \text{ flops}$$

There are 4 instances of core A.
 $= 675 \times 4 = 2700 \text{ flops.}$

CORE B:-

no. of int mode chains = 15 (core B core logic flops +
no. of ffs in each chain = 68 core B wrapper flops +
core A wrapper flops)

$$\text{total flops} = \frac{68}{\times 15} \\ \underline{\hspace{1cm}} 1020 \text{ flops}$$

$\therefore$ wrapper flops of core A are already included in the above 2700 flops.

$$= 1020 - \left[4 \times \left(\begin{array}{c} \text{no. of} \\ \text{core A ext mode} \times \text{length of} \\ \text{chains} \quad \text{each chain} \end{array} \right) \right]$$

$\therefore$ in ext mode, we have only wrappers

$$= 1020 - [4 \times (5 \times 27)]$$

$$= 1020 - [4 \times 135] = 480$$

$$\boxed{\text{Total ffs} = 480 + 2700 = 3180 \text{ ffs}}$$

FLOP COUNT

LEVEL 3 PROJECTS

FLOP COUNT

(contd...)

In level 4 projects there are around 44k flops. (106 scan chains each with 420 flops)

LEVEL 4 PROJECTS

Level 3

Functional clocks:

add_clocks clka -period 10ns

add_clocks clkb -period 13.2ns

add_clocks clkc -period 14ns

1/period will give you frequency

CLOCKS

Level 3

Scan clock :

freq- 25 MHz

Level 4

Functional clock

frequency : 400 MHz

Level 4

Scan clock

frequency : 25 MHz

CLOCKS

(contd...)

Design consists CoreB block inside which there are 4 instances of CoreA blocks.

CoreA:

int mode chains = 15

Length = 45 flops per chain

ext mode chains = 5

Length = 27 flops per chain

Compression ratio = 8x

CoreB:

int mode chains = 15

Length = 68 flops per chain

ext mode chains = 5

Length = 27 flops per chain

Compression ratio = 10x

LEVEL 3 PROJECTS

**Scan chain length and
compression ratio**

Level 4 projects :

No.of chains = 106

Length = 420 flops per chain

Compression ratio = 53x

LEVEL 4 PROJECTS

**Scan chain length and
compression ratio**

MBIST : Tessent MBIST

Scan Insertion : Tessent Scan

EDT : Tessent TestKompress

ATPG : Tessent TestKompress

Simulation : QuestaSim

TOOLS